

NORTH CAROLINA STATE UNIVERSITY
DEPARTMENT OF CHEMICAL AND BIOMOLECULAR ENGINEERING

CHE 435 / CHE 525 Process Systems Analysis and Control

Sample Midterm Exam 1 with Study Guide

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Name (print legibly): _____

- Test is closed book, closed notes. You may use an unlimited pages of torpedo sheet and a non-programmable calculator.
- Turn off all cell phones and smart watches and put them away. We will display a clock.
- Sit with a seat between each person if possible. Remove all hats, caps, and hoods. Keep your eyes on your own paper. To the extent you can, only display the page you are working on.
- If there is scrap work you don't want to be graded, write on the back of the paper or clearly cross it out. Only work in the boxed area will be graded.
- We will clarify wording, but we will not tell you how to do the problem or if your answer is correct. If you have a question, come up front to ask the instructor/TA.
- You can have additional paper if needed — raise your hand. **Any answers written on the back of the pages cannot be scanned and hence will not be graded.**
- Only one person can leave the room at a time. Leave your test in the room when you leave. No exits in the last 20 minutes of the test period.
- Before handing in the test, make sure the problems are in order and re-staple if necessary.
- Please do not discuss the exam within 48 hours, so as to keep the exam problems confidential until everyone has taken the exam.

1. Calculate the following.

(a) the modulus and argument of $i/(1+i)$.

Solution:

$$\left| \frac{i}{1+i} \right| = \frac{|i|}{|1+i|} = \frac{1}{\sqrt{2}}.$$

$$\angle \frac{i}{1+i} = \angle i - \angle(1+i) = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}.$$

(b) the Laplace transform of the following “pulse” signal:

$$f(t) = \begin{cases} 1/a, & 0 \leq t < a \\ 0, & t \geq a \end{cases}$$

where $a > 0$ is a constant. (Hint: To verify your answer, you may check that $F(s) \rightarrow 1$ as $a \rightarrow 0$.)

Solution:

$$F(s) = \int_0^\infty f(t)e^{-st} dt = \int_0^a \frac{1}{a} e^{-st} dt = \frac{1 - e^{-as}}{as}.$$

Alternatively, note that $f(t) = \frac{1}{a}(1 - S(t-a))$. For the delayed unit step $S(t-a)$, we have $\mathcal{L}[S(t-a)] = \frac{1}{s}e^{-as}$. Therefore,

$$F(s) = \frac{1}{a} \left(\frac{1}{s} - \frac{1}{s} e^{-as} \right) = \frac{1 - e^{-as}}{as}.$$

Indeed, as $a \rightarrow 0$ (the pulse approaches a unit impulse), $F(s) \rightarrow 1$. This is due to the limit $\lim_{x \rightarrow 0} \frac{1 - e^{-x}}{x} = 1$.

(c) the inverse Laplace transform of $1/(s^2 + 4)$.

Solution: We know that $\mathcal{L}[\sin 2t] = \frac{2}{s^2 + 4}$. Therefore,

$$\mathcal{L}^{-1} \left[\frac{1}{s^2 + 4} \right] = \frac{1}{2} \sin 2t.$$

Skills required:

- Use Euler’s formula to perform calculations on modulus and argument of complex numbers, or convert between forms of $e^{i\theta}$ and $\cos \theta + i \sin \theta$.
- Perform Laplace transform based on definition or common properties – linearity, multiplication by e^{-at} , multiplication by t , delayed signal, etc.
- Perform inverse Laplace transform based on the common properties, by memorization, derivation, and/or using a Laplace transform table.

2. Suppose that a process is described by the following ODE:

$$\frac{d^2y(t)}{dt^2} - 2\frac{dy(t)}{dt} + y(t) = 3u(t).$$

(a) What is the static gain of the system?

Solution: The transfer function is

$$\frac{Y(s)}{U(s)} = \frac{3}{s^2 - 2s + 1}.$$

Since the system is unstable (has poles not on the left half plane), there does not exist a static gain.

(b) Calculate and plot its *unit impulse* response.

Solution: For the unit impulse input, $U(s) = 1$ and hence $Y(s) = \frac{3}{(s-1)^2}$. By inverse Laplace transform,

$$y(t) = 3te^t.$$

(The plot is omitted.)

Skills required:

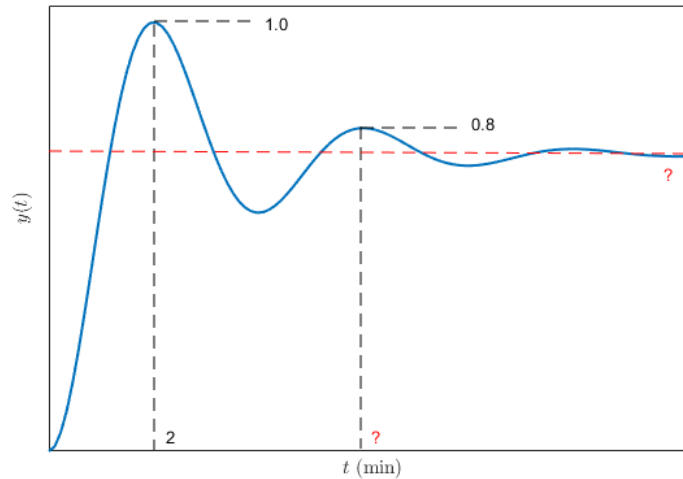
- Convert between an ODE and a transfer function model.
- Calculate impulse and step responses of a transfer function, by Laplace transform and its inverse.
- Indicate the poles, zeros, and static gain (but first whether they exist) of a transfer function.

Skills required:

- Convert between an ODE and a transfer function model.
- Calculate impulse and step responses of a transfer function, by Laplace transform and its inverse.
- Indicate the poles, zeros, and static gain (but first whether they exist) of a transfer function.

3. The *unit step* response of an unknown process is plotted below. The engineer John Doe suspects that it has a second-order transfer function

$$G(s) = \frac{K}{\tau^2 s^2 + 2\tau\zeta s + 1}.$$



- (a) If John is correct, when should the second peak occur?

Solution: The peak time $t_p = 2$ min, and hence the period is $2t_p = 4$ min. The second peak occurs at 6 min.

- (b) Determine the static gain K .

Solution: We use the relation that decay ratio = overshoot². That is,

$$\begin{aligned} \frac{0.8 - K}{1 - K} &= \left(\frac{1 - K}{K} \right)^2 \Rightarrow 0.8K^2 - K^3 = (1 - K)^3 \Rightarrow 0.8K^2 - K^3 = 1 - 3K + 3K^2 - K^3 \\ &\Rightarrow 2.2K^2 - 3K + 1 = 0 \Rightarrow K = 0.7834 \end{aligned}$$

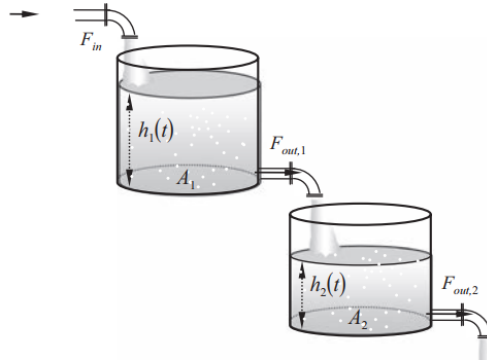
Hint: For a second-order system, time to first peak $t_p = \frac{\pi\tau}{\sqrt{1-\zeta^2}}$, period = $2\pi\tau/\sqrt{1-\zeta^2} = 2t_p$, overshoot (as a ratio) = $\exp(-\frac{\pi\zeta}{\sqrt{1-\zeta^2}})$, decay ratio = $\exp(-\frac{2\pi\zeta}{\sqrt{1-\zeta^2}})$ = overshoot².

Disclaimer: Ruler measurements on the paper may not represent the actual relations.

Skills required:

- For first- and underdamped second-order systems, find the characteristic quantities of the step response from the parameters (K, τ, ζ).
- For first- and underdamped second-order systems, find the system parameters from the data on the step response curve.
- For second-order systems, discriminate overdamped, critically damped, and underdamped systems, and explain the difference of their qualitative behaviors.

4. A system of two liquid storage tanks in series is shown in the figure below.



- (a) Assume that the volumetric flow rate of the outgoing stream from each tank is proportional to the square root of liquid height ($F_{out,1} = C_1 \sqrt{h_1(t)}$ and $F_{out,2}(t) = C_2 \sqrt{h_2(t)}$). Use material balance equations to derive a dynamic model for the system of tanks.

Solution: The model should be

$$\frac{dh_1}{dt} = \frac{F_{in}(t) - C_1 \sqrt{h_1(t)}}{A_1}, \quad \frac{dh_2}{dt} = \frac{C_1 \sqrt{h_1(t)} - C_2 \sqrt{h_2(t)}}{A_2}.$$

- (b) At a steady state, the inlet volumetric flow rate is $\bar{F}_{in}$. Express the liquid heights in the two tanks, $\bar{h}_1$ and $\bar{h}_2$, in terms of $\bar{F}_{in}$, and then linearize the dynamic model at this steady state.

Solution: At the steady state, $dh_1/dt = 0$ and $dh_2/dt = 0$. Hence,

$$\bar{F}_{in} - C_1 \sqrt{\bar{h}_1} = 0, \quad C_1 \sqrt{\bar{h}_1} - C_2 \sqrt{\bar{h}_2} = 0.$$

That is,

$$\bar{h}_1 = (\bar{F}_{in}/C_1)^2, \quad \bar{h}_2 = (C_1/C_2)^2 \bar{h}_1 = (\bar{F}_{in}/C_2)^2.$$

Given the partial derivatives:

$$\begin{aligned} \frac{\partial}{\partial h_1} \left(\frac{F_{in} - C_1 \sqrt{h_1}}{A_1} \right) &= -\frac{C_1}{2A_1 \sqrt{h_1}} \xrightarrow{\text{steady state}} -\frac{C_1}{2A_1 (\bar{F}_{in}/C_1)} = -\frac{C_1^2}{2A_1 \bar{F}_{in}} \\ \frac{\partial}{\partial F_{in}} \left(\frac{F_{in} - C_1 \sqrt{h_1}}{A_1} \right) &= \frac{1}{A_1} \xrightarrow{\text{steady state}} \frac{1}{A_1} \\ \frac{\partial}{\partial h_1} \left(\frac{C_1 \sqrt{h_1} - C_2 \sqrt{h_2}}{A_2} \right) &= \frac{C_1}{2A_2 \sqrt{h_1}} \xrightarrow{\text{steady state}} \frac{C_1}{2A_2 (\bar{F}_{in}/C_1)} = \frac{C_1^2}{2A_2 \bar{F}_{in}} \\ \frac{\partial}{\partial h_2} \left(\frac{C_1 \sqrt{h_1} - C_2 \sqrt{h_2}}{A_2} \right) &= -\frac{C_2}{2A_2 \sqrt{h_2}} \xrightarrow{\text{steady state}} -\frac{C_2}{2A_2 (\bar{F}_{in}/C_2)} = -\frac{C_2^2}{2A_2 \bar{F}_{in}} \end{aligned}$$

the linearized model is

$$\frac{dh'_1}{dt} = \frac{1}{A_1} F'_{in}(t) - \frac{C_1^2}{2A_1 \bar{F}_{in}} h'_1(t), \quad \frac{dh'_2}{dt} = \frac{C_1^2}{2A_2 \bar{F}_{in}} h'_1(t) - \frac{C_2^2}{2A_2 \bar{F}_{in}} h'_2(t).$$

- (c) If a volume V_0 of liquid is suddenly pushed into the first tank (i.e., $F'_{in}(t) = V_0\delta(t)$), how will $h_2(t)$ respond?

Solution: By Laplace transform, we find the following transfer functions of the two tanks, respectively:

$$\frac{H'_1(s)}{F'_{in}(s)} = \frac{\frac{1}{A_1}}{s + \frac{C_1^2}{2A_1\bar{F}_{in}}} = \frac{\frac{2\bar{F}_{in}}{C_1^2}}{\frac{2A_1\bar{F}_{in}}{C_1^2}s + 1}, \quad \frac{H'_2(s)}{H'_1(s)} = \frac{\frac{C_1^2}{2A_2\bar{F}_{in}}}{s + \frac{C_2^2}{2A_2\bar{F}_{in}}} = \frac{\frac{C_1^2}{C_2^2}}{\frac{2A_2\bar{F}_{in}}{C_2^2}s + 1}.$$

Connecting the two tanks (i.e., multiplying the two transfer functions), we obtain

$$\frac{H'_2(s)}{F'_{in}(s)} = \frac{\frac{C_2^2}{2A_2\bar{F}_{in}}}{s + \frac{C_2^2}{2A_2\bar{F}_{in}}} = \frac{\frac{C_1^2}{C_2^2}}{\left(\frac{2A_1\bar{F}_{in}}{C_1^2}s + 1\right)\left(\frac{2A_2\bar{F}_{in}}{C_2^2}s + 1\right)}.$$

This is a second-order system with $K = C_1^2/C_2^2$.

- If $A_1/C_1^2 = A_2/C_2^2$, denote $\tau = 2A_1\bar{F}_{in}/C_1^2$. The system is critically damped: $G(s) = K/(\tau s + 1)^2$. Given that $F'_{in}(s) = V_0$, we have $H'_2(s) = KV_0/(\tau s + 1)^2$. By inverse Laplace transform, we have

$$h'_2(t) = \frac{KV_0 t}{\tau^2} e^{-t/\tau}.$$

- If $A_1/C_1^2 \neq A_2/C_2^2$, denote $\tau_1 = 2A_1\bar{F}_{in}/C_1^2$ and $\tau_2 = 2A_2\bar{F}_{in}/C_2^2$. The system is over-damped: $G(s) = K/(\tau_1 s + 1)(\tau_2 s + 1)$. We have

$$H'_2(s) = \frac{KV_0}{(\tau_1 s + 1)(\tau_2 s + 1)} = \frac{KV_0}{\tau_1 \tau_2} \frac{1}{s + \frac{1}{\tau_1}} \frac{1}{s + \frac{1}{\tau_2}} = \frac{KV_0}{\tau_1 \tau_2} \frac{1}{\frac{1}{\tau_2} - \frac{1}{\tau_1}} \left(\frac{1}{s + \frac{1}{\tau_1}} - \frac{1}{s + \frac{1}{\tau_2}} \right).$$

By inverse Laplace transform,

$$h'_2(t) = \frac{KV_0}{\tau_1 - \tau_2} \left(e^{-t/\tau_1} - e^{-t/\tau_2} \right).$$

Skills required:

- Use material and energy balance to build the model of a plant.
- Indicate the inputs (cause) and outputs (outcome) based on the process description and engineering common sense.
- Explain the meaning of a steady state (d/dt of every variable is 0), and determine the input-output relation at the steady state.
- Linearize the process model around the steady state through partial differentiation.
- Determine the transfer function in the presence of parallel or series connections of transfer functions.
- Perform calculations in symbols.